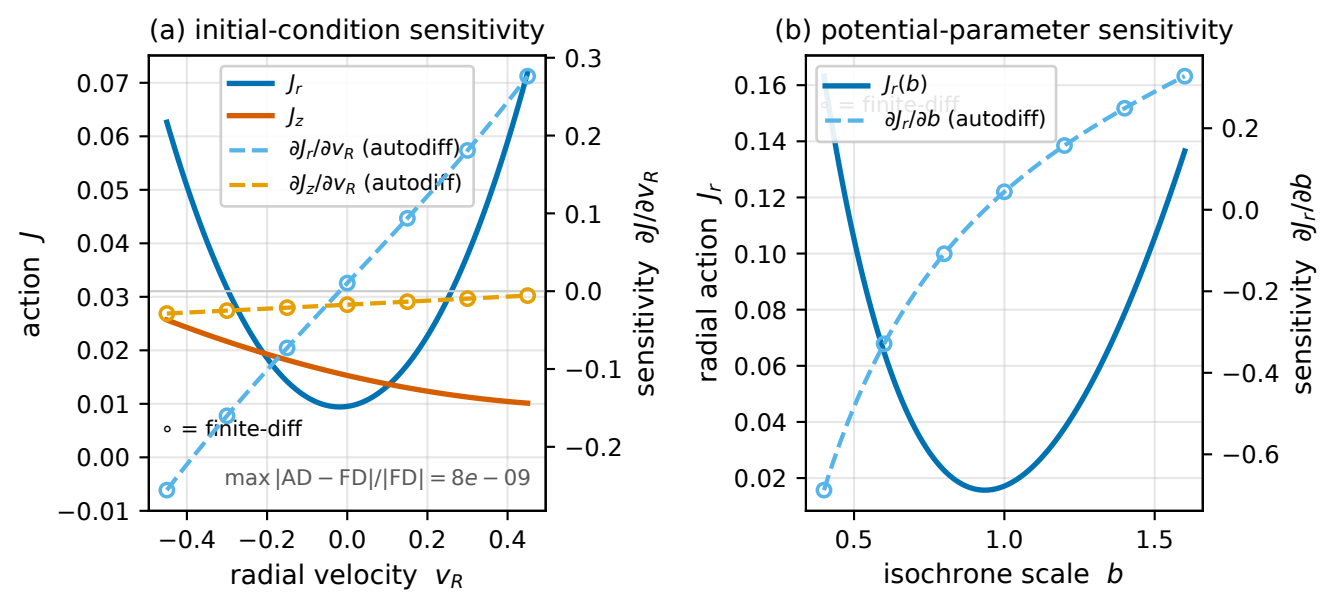
(a) initial-condition sensitivity
J_r
J_z
$\partial J_r/\partial v_R$ (autodiff)
$\partial J_z/\partial v_R$ (autodiff)
∘ = finite-diff
max |AD − FD|/|FD| = 8e − 09
action J
sensitivity $\partial J/\partial v_R$
radial velocity v_R
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.2
0.0
0.2
0.4
(b) potential-parameter sensitivity
$J_r(b)$
$\partial J_r/\partial b$ (autodiff)
radial action J_r
sensitivity $\partial J_r/\partial b$
isochrone scale b
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.2
0.0
−0.2
−0.4
−0.6
0.5
1.0
1.5